Robot Motion

Probabilistic Robotics, chapter 5

Lesson t2.4 · Thrun, Burgard, Fox

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Why a robot needs a probabilistic motion model

- A filter's predict step is **$p(x_t \mid u_t, x_{t-1})$** (5.3): where is the robot after a command?
- Kinematics gives one pose; real wheels, tracks and floors give a **cloud** of poses
- Cheapest fix: keep the kinematics, put **noise on what drives them**
- Models that overestimate uncertainty work better than tight ones (p. 118)
- Result: a tracked robot, a wheeled base or a walking robot can all use the same two models

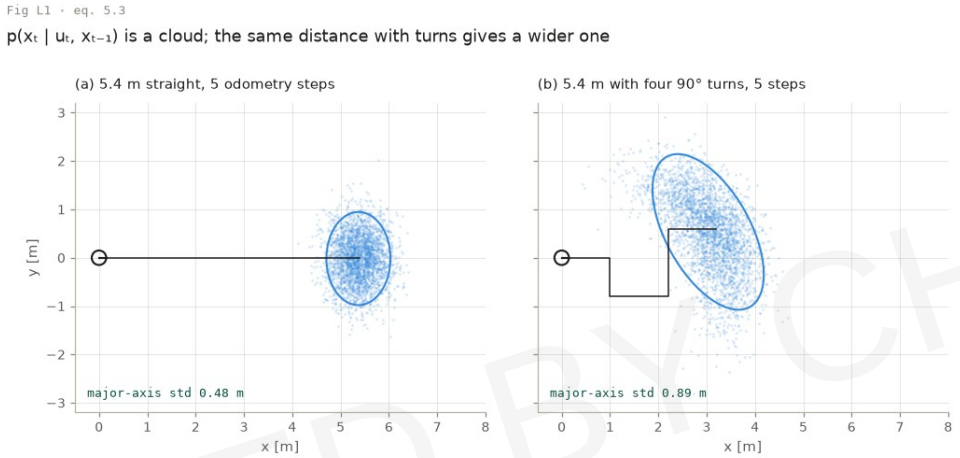
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Roadmap

1. Motion as a distribution: two uses of a motion model (5.1, 5.2.2)
2. The velocity model: circle, noise on (v, ω), the extra rotation $\hat{\gamma}$ (5.3)
3. Error densities and samplers: normal and triangular (Tables 5.2, 5.4)
4. The odometry model: rotate, translate, rotate (5.4)
5. Conditioning on a map, and what it gets wrong (5.5)

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A motion gives a cloud of poses



- Straight path: major-axis standard deviation 0.481 m
- Path with four 90° turns: 0.888 m
- Every turn's heading error swings what follows

Same distance driven straight or with turns, same noise

Same distance, nearly double the spread: heading error drives the cloud

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Two models, two uses

Velocity model

- Input u = (v, ω): the command
- Works before the motion happens
- Planners use it

Odometry model

- Input: two odometry poses
- Only exists after the motion
- More accurate, so estimators use it (p. 121)

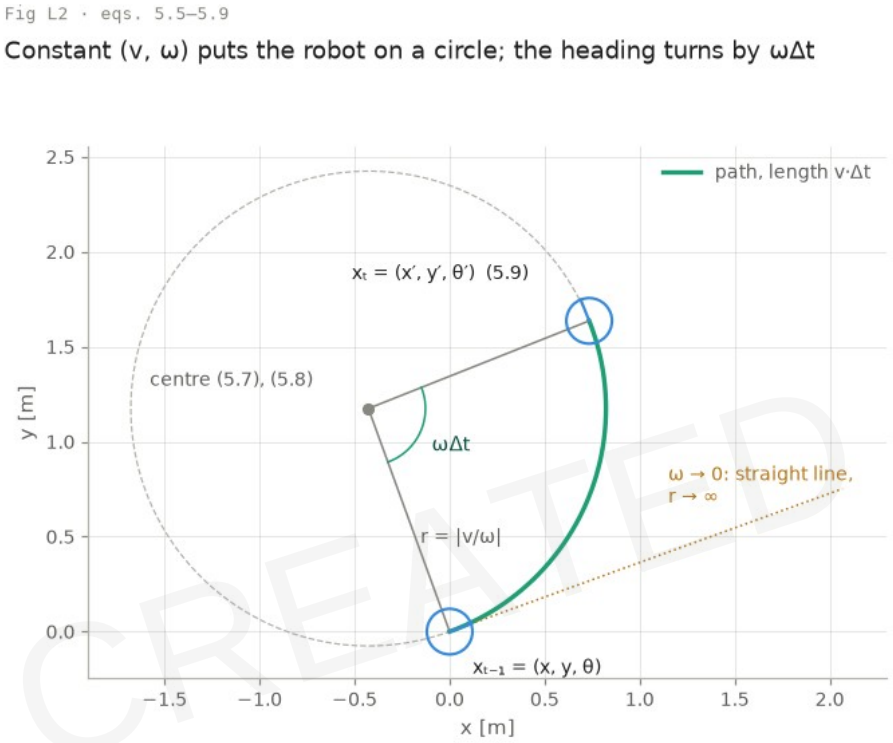
Both

- Pose is (x, y, θ) (5.1); θ = 0 along x, counterclockwise positive
- θ lives on a circle: **wrap** every angle difference

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Planners predict from commands; estimators look back at odometry

Exact motion is a circle



- Radius r = |v/ω| about a centre beside the robot
- Heading turns by exactly the angle swept
- ω = 0: the centre is at infinity

Constant (v, ω) over Δt

A constant command moves a differential-drive base along an exact arc

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Motion on the circle

$$x' = x - \frac{v}{\omega}\sin\theta + \frac{v}{\omega}\sin(\theta + \omega\Delta t) \tag{5.9}$$

$$y' = y + \frac{v}{\omega}\cos\theta - \frac{v}{\omega}\cos(\theta + \omega\Delta t) \tag{5.9}$$

$$\theta' = \theta + \omega\Delta t \tag{5.9}$$

(x, y, θ) pose before the move

(x′, y′, θ′) pose after Δt

v, ω forward speed, turn rate

v/ω radius of the circle

The pose after Δt is a point on a circle about the centre (5.7), (5.8)

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Predict: noise on v and ω only

- Perturb only v and ω, run (5.9), many times

? **Where do the samples fill in (x, y, θ), and can a sensor suggest a pose off that set?**

A A 3-D blob; any nearby pose is possible

B A 2-D sheet; a pose off it has probability zero

C A line; only distance varies

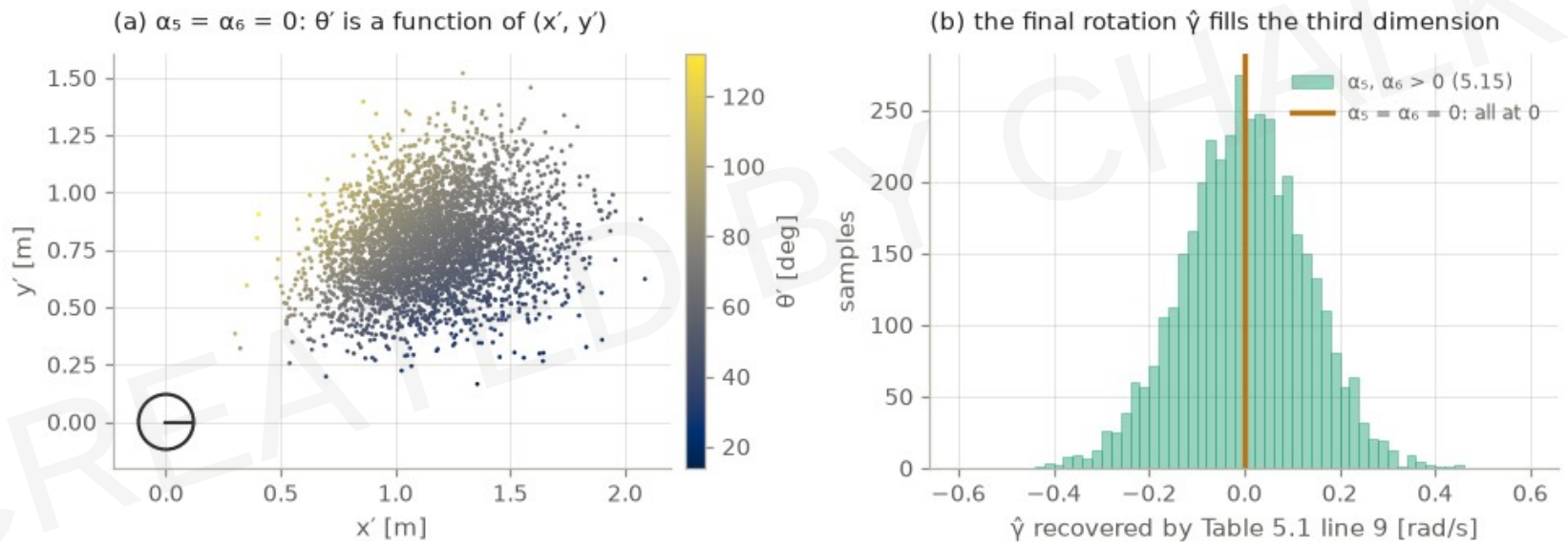
D A 2-D sheet; sensors can still suggest off-sheet poses

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The final rotation ŷ gives the cloud thickness

Fig L3 · eqs. 5.13–5.16

With two noise terms every sample lies on one surface; ŷ thickens it



Samples coloured by final heading θ′

Without ŷ the cloud is a sheet; with it the cloud has thickness

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Noisy controls and the final rotation

$$\hat{v} = v + \varepsilon_{\alpha_1 v^2 + \alpha_2 \omega^2}, \quad \hat{\omega} = \omega + \varepsilon_{\alpha_3 v^2 + \alpha_4 \omega^2} \tag{5.10}$$

$$\theta' = \theta + \hat{\omega}\Delta t + \hat{\gamma}\Delta t \tag{5.14}$$

$$\hat{\gamma} = \varepsilon_{\alpha_5 v^2 + \alpha_6 \omega^2} \tag{5.15}$$

v̂, ω̂ noisy speeds the robot really executes

γ̂ extra final rotation, a modelling device

ε_b² zero-mean noise with variance b^2

α_1–α_6 robot-specific noise parameters

Noise grows with commanded speed: a robot at rest does not drift

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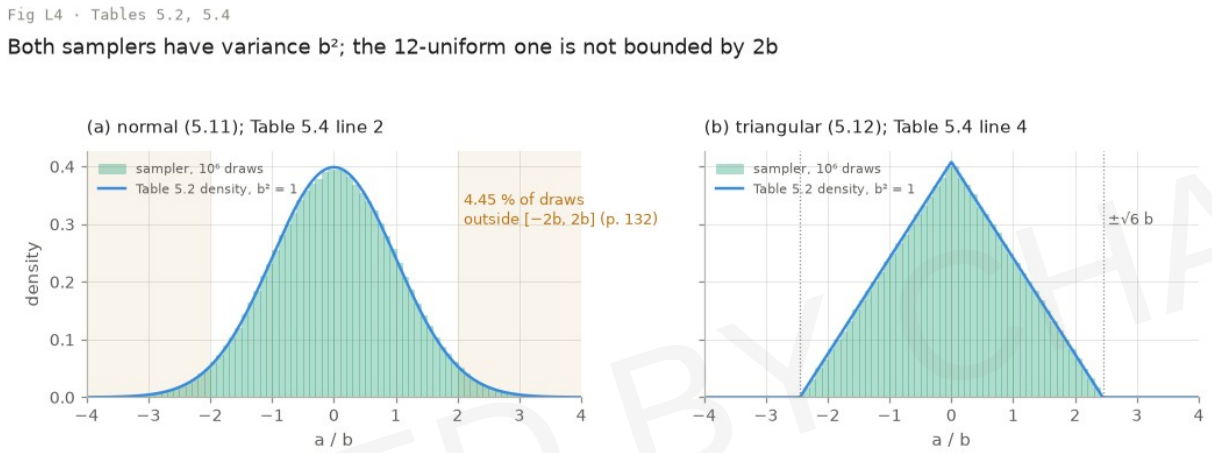
Why the third noise term exists

Two noise variables cannot fill three pose dimensions, so add $\hat{\gamma}$

Models that overestimate uncertainty work better than tight ones (p. 118)

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Error densities: normal and triangular



- Normal: infinite support
- Triangular: zero beyond ±√6 b
- 12-uniform sampler: not bounded by 2b

Both have variance b²

Same variance b², very different tails: a zero density never recovers

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Table 5.2: the two error densities

$$p_{\text{normal}}(a, b^2) = \frac{1}{\sqrt{2\pi b^2}} \exp\left(-\frac{a^2}{2b^2}\right) \tag{5.11}$$

$$p_{\text{tri}}(a, b^2) = \max\left(0, \frac{1}{\sqrt{6}\, b} - \frac{|a|}{6b^2}\right) \tag{5.12}$$

a error to score or draw

b² variance of the error

p_normal infinite support

p_tri zero beyond ±√6 b

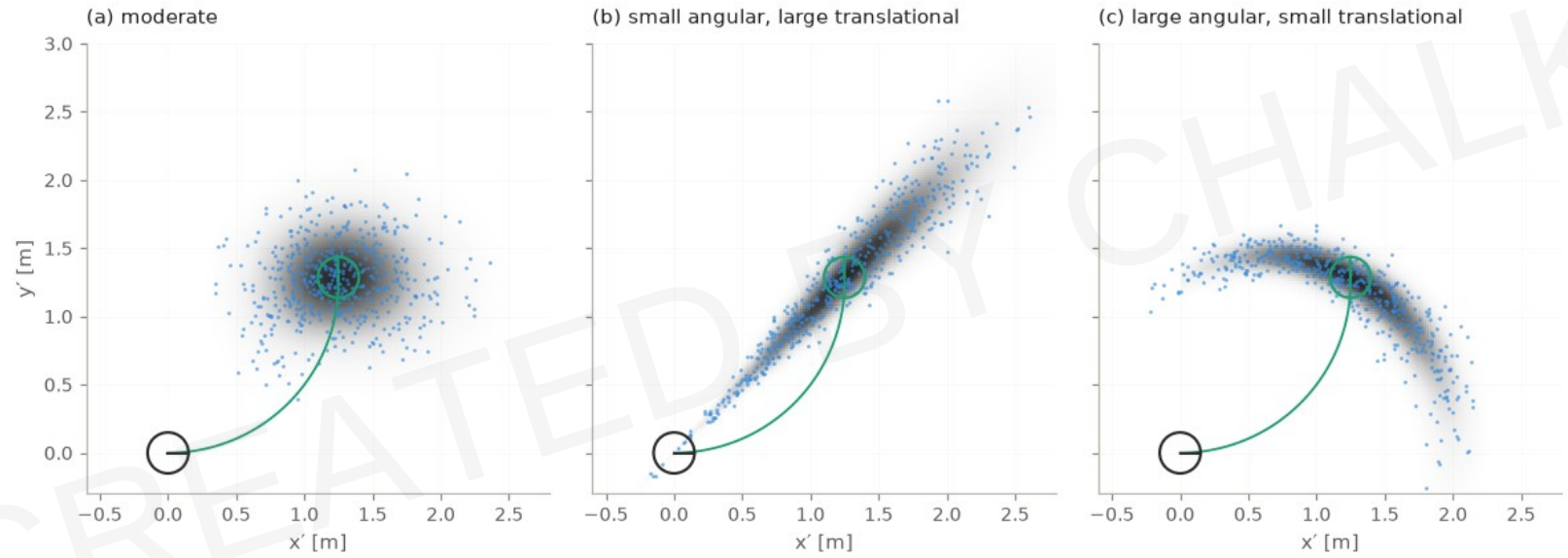
At rest b² = 0 and the normal is 0/0: put a floor on b²

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Density and sampler for the velocity model

Fig L5 · Tables 5.1, 5.3

Density (gray) and 500 samples (blue) agree; v-noise stretches the cloud along a ray, ω-noise bends it



- Noise on v stretches the cloud along a ray
- Noise on ω bends it into a banana

Gray: Table 5.1 density summed over θ′. Blue: Table 5.3 draws

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Sampling the velocity model, Table 5.3

1 Perturb controls

Draw $\hat{v}$, $\hat{\omega}$ and $\hat{\gamma}$ with variances from α_1–α_6 (5.10), (5.15); lines 2–4

2 Run the circle

Apply the arc of (5.9) with $\hat{v}$ and $\hat{\omega}$ (5.16); lines 5–6

3 Turn the heading

$\theta' = \theta + (\hat{\omega} + \hat{\gamma})\Delta t$ (5.14); line 7

- Lines 5–6 divide by $\hat{\omega}$: a straight move needs its own branch

The sampler is a simulator: perturb the controls, run the kinematics forward

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Three traps in Table 5.1 and Table 5.3

- **Line 7 drops the sign of $\hat{v}$**: r^* is positive, so $\hat{v}$ takes the sign of the turn
- **Straight line**: Table 5.1 line 2 and Table 5.3 lines 5–6 divide by zero
- **Lines 6 and 9 are angle differences**: wrap them
- Eq. (5.32) is a density over noise variables, not over poses

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Predict: a right turn through Table 5.1

- Command $(v, \omega) = (1, -0.5)$, $\Delta t = 1$ s
- Density asked at the noise-free end pose

? **What does line 7 give for $\hat{v}$, and how likely is the true end pose?**

A $\hat{v} = +1$; the true pose is most likely

B $\hat{v} = -1$; the true pose is nearly impossible

C $\hat{v} = 0$; the density is undefined

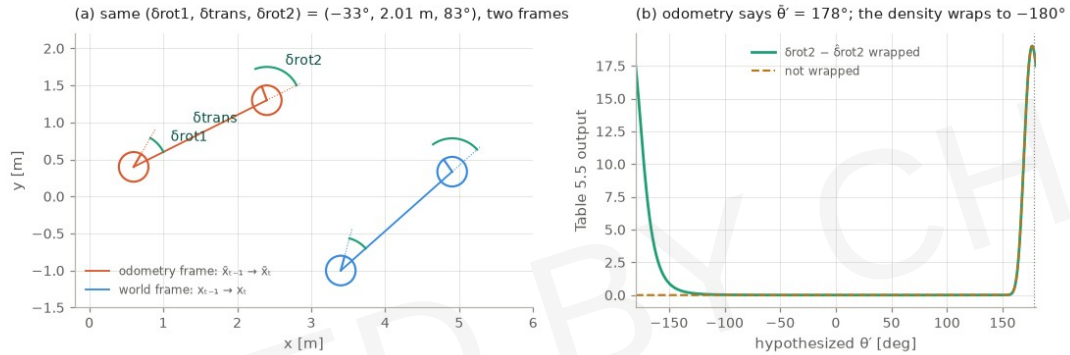
D $\hat{v} = -0.5$; the pose is slightly unlikely

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Odometry: rotate, translate, rotate

Fig L6 · Table 5.5, eqs. 5.34–5.36

Odometry is used only as a relative motion; its angles live on a circle



- Odometry frame drifts; its differences do not
- (δrot1, δtrans, δrot2) is a sufficient statistic (p. 135)
- Example: (−33.4°, 2.012 m, 83.4°)

The same three numbers move the robot in the world frame

Three numbers (δrot1, δtrans, δrot2) carry the whole relative motion

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Decomposing the odometry pair

$$\delta_{\text{rot1}} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta} \tag{5.34}$$

$$\delta_{\text{trans}} = \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2} \tag{5.35}$$

$$\delta_{\text{rot2}} = \bar{\theta}' - \bar{\theta} - \delta_{\text{rot1}} \tag{5.36}$$

u pair of odometry poses (5.33)

x̄, ȳ, θ̄ odometry pose before

x̄′, ȳ′, θ̄′ odometry pose after

δrot1, δrot2 first and second rotation

δtrans straight distance

Rotate toward the target, drive straight, rotate to the final heading

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Traps in the odometry model

- **Wrap every angular difference**: p. 135 calls the bug hard to debug
- At the ±180° seam, 47.7 % of the density over θ' is lost without wrapping
- **Rotation in place**: $\delta_{trans} = 0$ makes atan2(0, 0) arbitrary
- **Table 5.5 vs (5.44)–(5.46)**: the book disagrees with itself on $\hat{\delta}$ or δ

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Dead reckoning: uncertainty only grows

Fig L7 · Table 5.6, Fig 5.10

Dead reckoning: the cloud only grows, and each turn's heading error swings it sideways

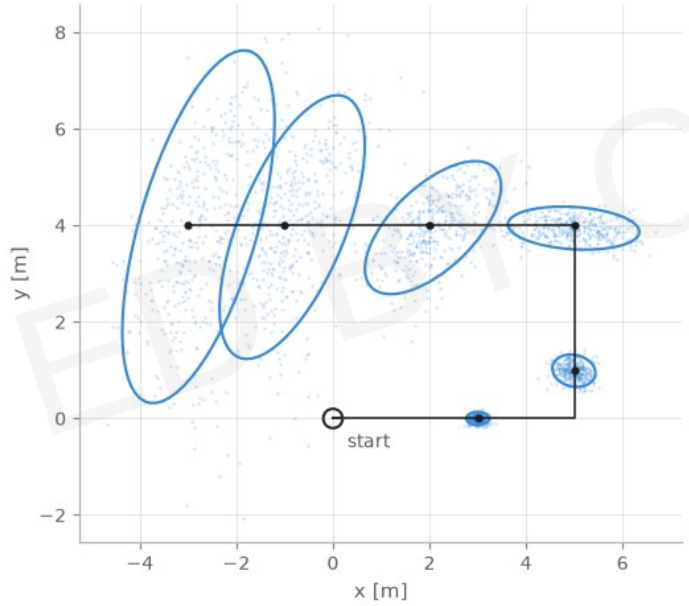


Table 5.6 sampled step by step around a rectangle

Nothing in the motion model shrinks the cloud; only a measurement can

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Dead reckoning: spread along the path

0.118 m

after 3 m

0.888 m

after 12 m

1.894 m

at the end of 17.0 m

- Major-axis standard deviation, Table 5.6 sampled with no measurements

Spread grows faster than linearly: heading error acts on every later metre

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Sampling the odometry model, Table 5.6

1 Recover

Turn the odometry pair into δrot1, δtrans, δrot2 (5.34)–(5.36); lines 2–4

2 Perturb

Subtract noise from each δ, variances scaled by the measured δ (5.37)–(5.39)

3 Compose

Apply the noisy $\hat{\delta}$ to the particle's own pose (5.40); lines 8–10

- Each particle moves from its own θ, not the odometer's

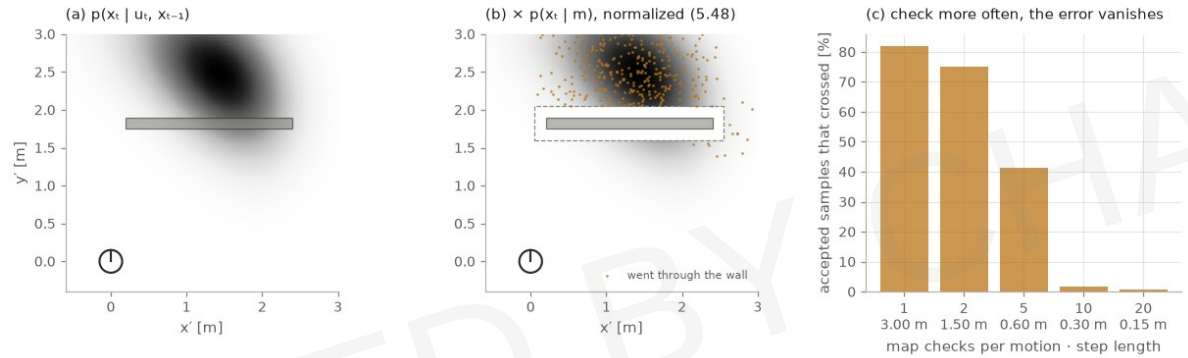
Noise enters in the three motion parameters, not in x and y

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A map removes mass, but only at the end pose

Fig L8 · eq. 5.48, Table 5.7

Checking only the final pose lets the robot through the wall



- 82.1 % of samples land in free space
- One map check per step: 81.9 % of those crossed the wall
- Ten checks (0.3 m steps): 1.6 %

Mass inside the grown obstacle is removed; mass behind it stays

Checking only the end pose lets particles tunnel through walls

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Conditioning on the map

$$p(x_t \mid u_t, x_{t-1}, m) = \eta \frac{p(x_t \mid u_t, x_{t-1})\, p(x_t \mid m)}{p(x_t)} \tag{5.48}$$

- **η** normalizer
- **m** map
- **$p(x_t \mid m)$** zero in occupied space, constant elsewhere
- **$p(x_t)$** pose prior, taken as uniform

- Sampler: draw until $p(x_t \mid m) > 0$ (Table 5.7); **cap the retries**

Table 5.7 loops until the sample is free: cap the retries

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Symptom to fix

Symptom	Cause	Fix
Filter confident and wrong	α too small	Inflate α (p. 118)
Density zero off the sheet	$\alpha_5 = \alpha_6 = 0$	Final-rotation noise (5.15)
NaN at rest	$b^2 = 0$ when $v = \omega = 0$	Floor on b^2
NaN on straight lines	$\omega = 0$ in Table 5.1 or 5.3	Straight-line branch
Right turns unlikely	Line 7 loses sign of $\hat{v}$	Sign from displacement
Likelihood collapses at ±180°	Unwrapped difference	Wrap every angle
Particles behind a wall	Map checked at end pose	Shorter steps, more checks
Map sampler hangs	Unbounded do–until	Cap retries, weight 0

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Summary: claims you should be able to defend

- A motion model is kinematics plus noise on the **controls**, not on the pose
- Two noise terms leave a 2-D sheet in pose space; the third term, $\hat{\gamma}$, fills it
- Every algorithm has a **density form** and a **sampling form**
- Wrap every angular difference; floor every variance
- Dead reckoning's uncertainty only grows; only measurements shrink it
- A map checked at the end pose lets particles through walls

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What comes next

- Chapter 6: the **update** half; same lesson, the measurement model
- Chapter 7: linearize the velocity model for EKF and UKF localization
- Chapter 8: the samplers become line 4 of Monte Carlo localization
- Chapter 13: the same samplers, per particle, in FastSLAM
- Exercises 1–3 (a robot with dynamics) are the first step toward drones

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